

WASP-12b

R-2015

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_220113 · created 2026-08-07T23:32:19+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459592.80819 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.044
Transit depth (Rp/Rs)^2	2.4 +/- 1.38 [%]
Orbital Inclination (inc)	77.36 +/- 2.86
Airmass coefficient 1 (a1)	0.25 +/- 0.15
Airmass coefficient 2 (a2)	1.0 +/- 0.62
Scatter in the residuals of the lightcurve fit is	49.37 %
Best Comparison Star	#2 - [125, 197]
Optimal Aperture	0.87
Optimal Annulus	3.77
Transit Duration (day)	0.0489 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.044 vs 0.1178 (deviation 0.85σ)
Duration fit vs published	0.0489 d vs 0.125 d (deviation 10.57σ)
Mid-time O-C	-1.1 min
Depth SNR	3.5
Residual scatter	49.37 %

Light curve

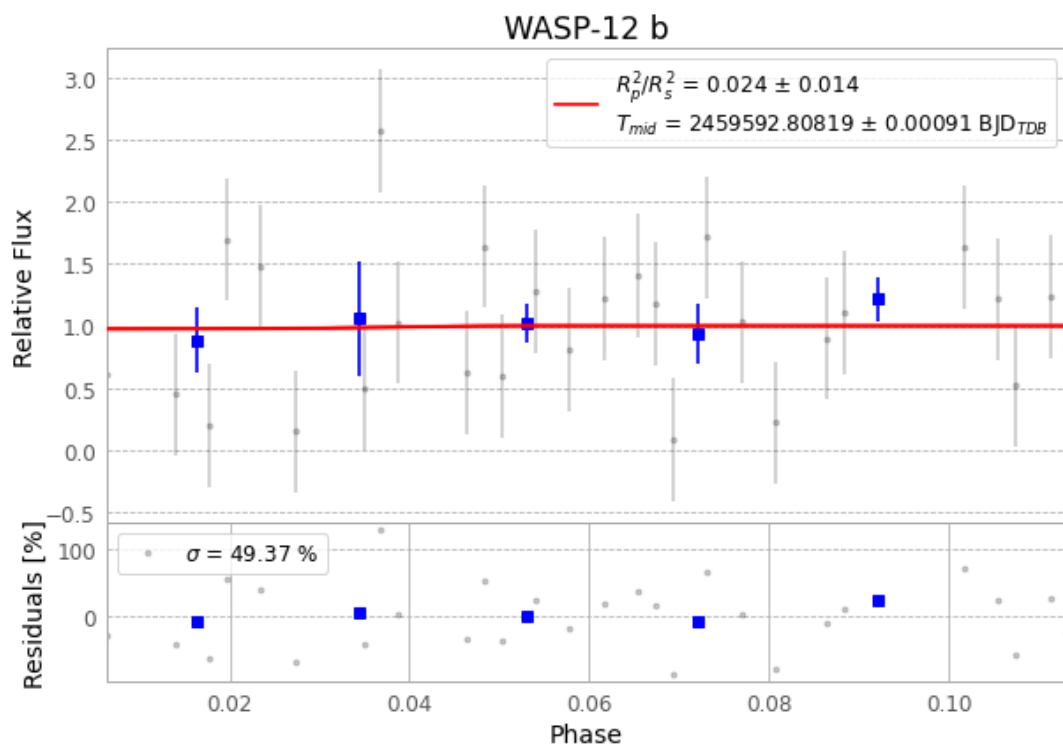


Figure 1 — FinalLightCurve_WASP-12 b_2022-01-13.png (SHA-256 017daffda3666603...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [439, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f5e303293c31f819...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

57 FITS frames (37 MB), WASP-12220113072412.FITS → WASP-12220113101213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-220113.log (b808393e5299...), FinalLightCurve_WASP-12 b_2022-01-13.png (017daffda366...), FinalLightCurve_WASP-12 b_2022-01-13.csv (0e85a47c57c2...)

Compute resources

Wall time 111.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 23:32Z.